

WASP-10b

R-0672

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_251006 · created 2026-07-27T18:41:32+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460954.6496 +/- 0.0051 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.212 +/- 0.014
Transit depth (Rp/Rs)^2	4.48 +/- 0.59 [%]
Orbital Inclination (inc)	86.32 +/- 1.15
Airmass coefficient 1 (a1)	1.222 +/- 0.034
Airmass coefficient 2 (a2)	-0.188 +/- 0.057
Scatter in the residuals of the lightcurve fit is	2.77 %
Best Comparison Star	None
Optimal Aperture	2.72
Optimal Annulus	8.18
Transit Duration (day)	0.075 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.212 $\pm$ 0.014 vs 0.1592 (deviation 3.31 σ)
Duration fit vs published	0.075 d vs 0.0946 d (deviation 1.01 σ)
Mid-time O–C	4.2 min
Depth SNR	13.3
Residual scatter	2.77 %

Light curve

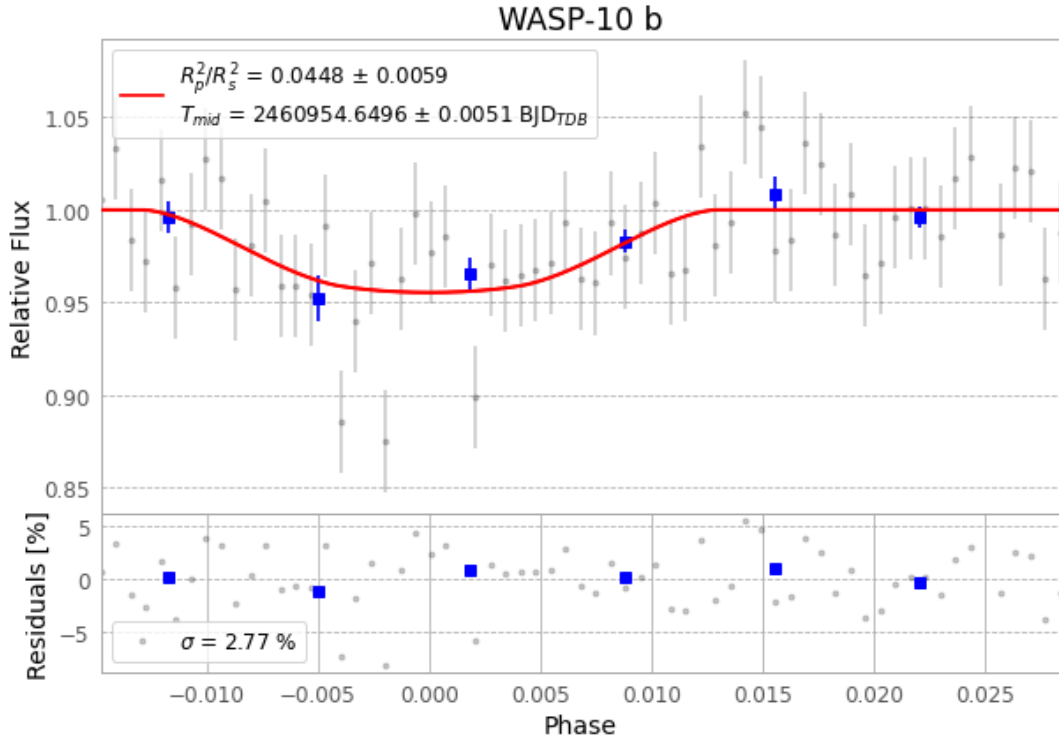


Figure 1 — FinalLightCurve_WASP-10 b_2025-10-05.png (SHA-256 66899bc3ae9f79a0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [369, 238], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 60461a03e9c57c74...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62f7813

Data provenance

65 FITS frames (43 MB), WASP-10251006022115.FITS → WASP-10251006053316.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-251006.log (45c094fa2e29...), FinalLightCurve_WASP-10 b_2025-10-05.png (66899bc3ae9f...), FinalLightCurve_WASP-10 b_2025-10-05.csv (92820bcdc59d...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 18:41Z.